

LAB 1

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Deadline: 2026/09/27 (Sun) 23:59

Demo: 2026/10/06 (Tue)

In this lab,

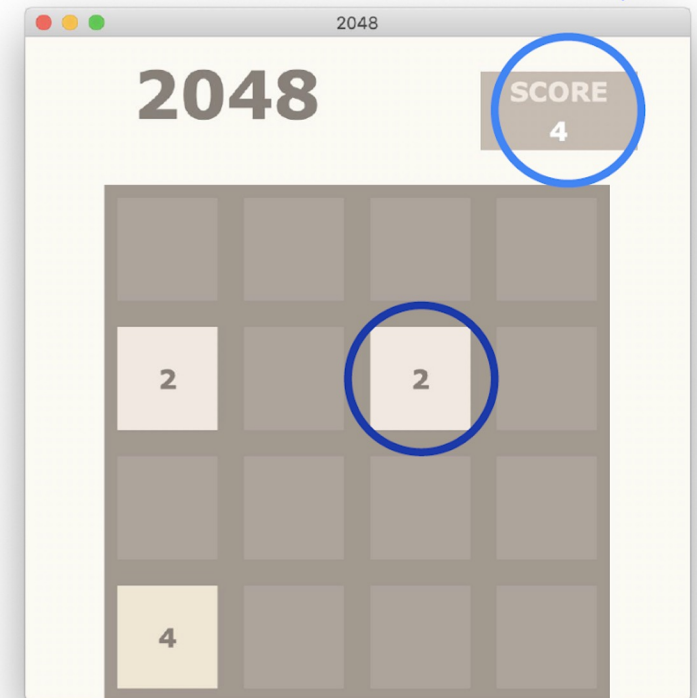
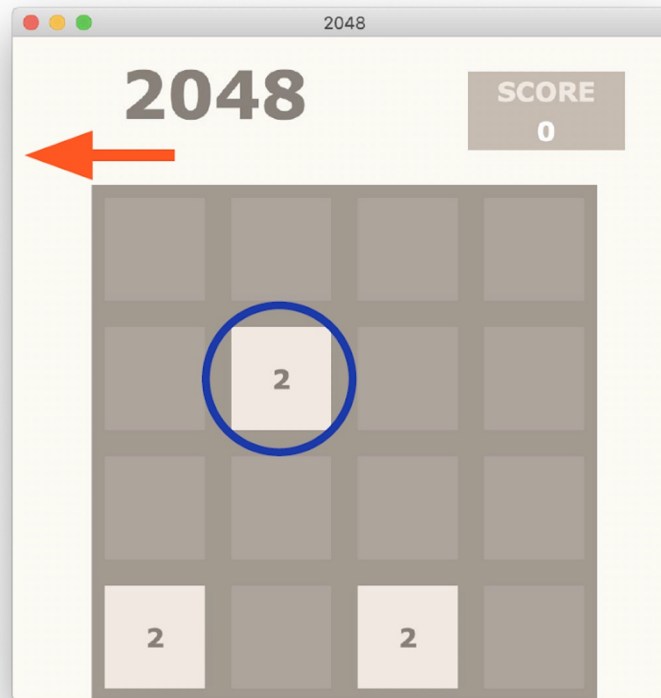
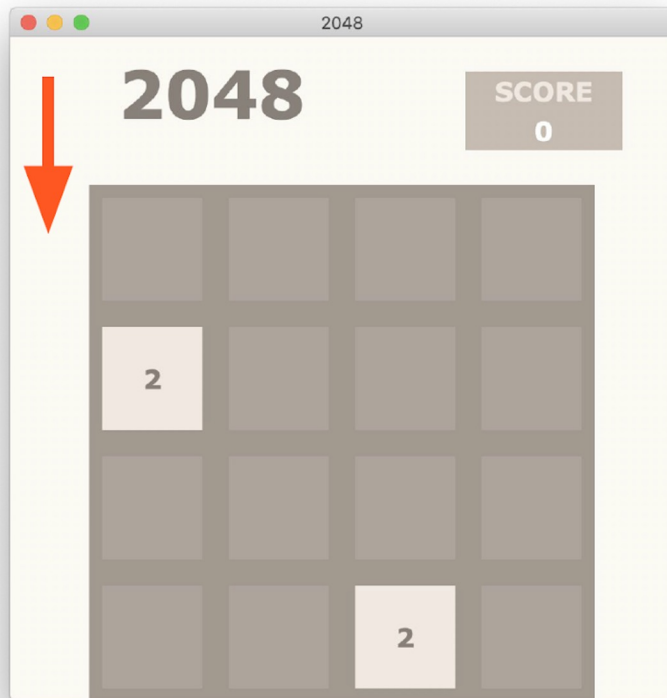
**Must use sample code,
otherwise no credit.**

Outline

- **2048 Game Rule**
- **Game State**
- **Temporal Difference Learning**
- **n-tuple Network**
- **Sample Code**
- **Scoring Criteria**
- **Reminders**

2048 Game Rules

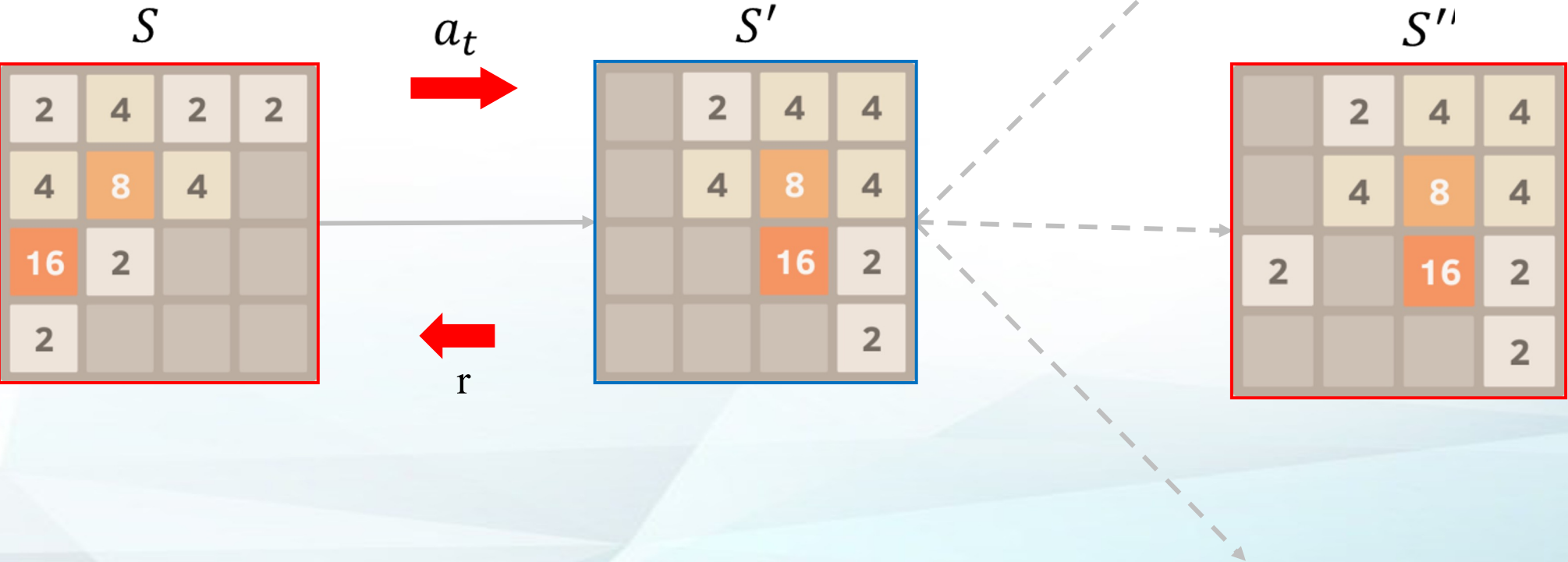
popup: **2** (90%), **4** (10%)



Game State

 *beforestate*
 *afterstate*

 perform action
 popup a random tile



Temporal Difference Learning (TD)

For each episode,

```
Initialize (before-)state  $s$ 
```

```
While  $s$  is not terminal do
```

```
     $a \leftarrow \operatorname{argmax}_a \text{EVALUATE}(s, a')$ 
```

```
     $r, s', s'' \leftarrow \text{MAKE\_MOVE}(s, a)$ 
```

```
    STORE( $s, a, r, s', s''$ )
```

```
     $s \leftarrow s''$ 
```

```
End While
```

```
For ( $s, a, r, s', s''$ ) from terminal down to initial do
```

```
    LEARN_EVALUATION( $s, a, r, s', s''$ )
```

```
End For
```

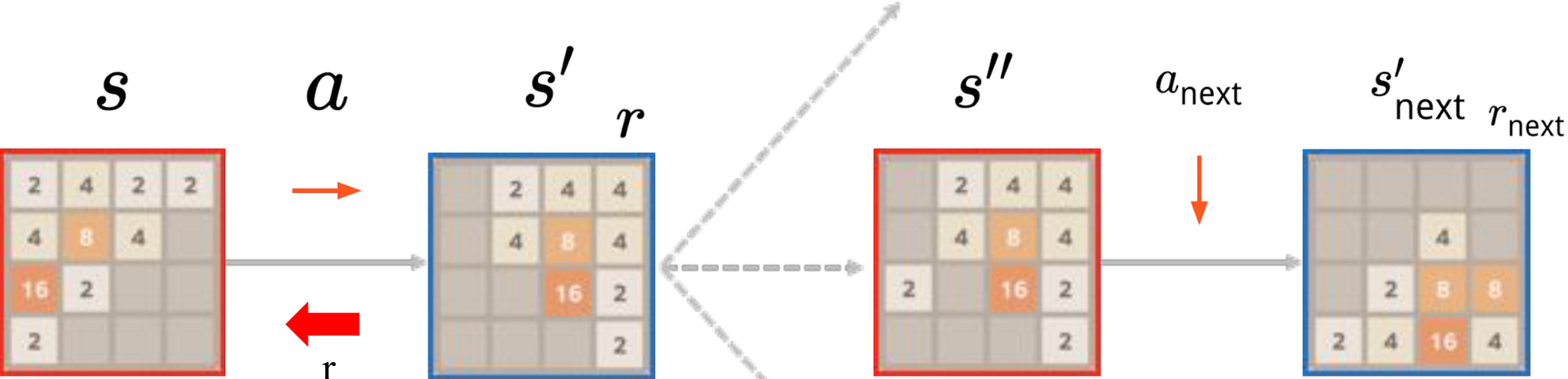


perform TD backup

TD Backup Diagram

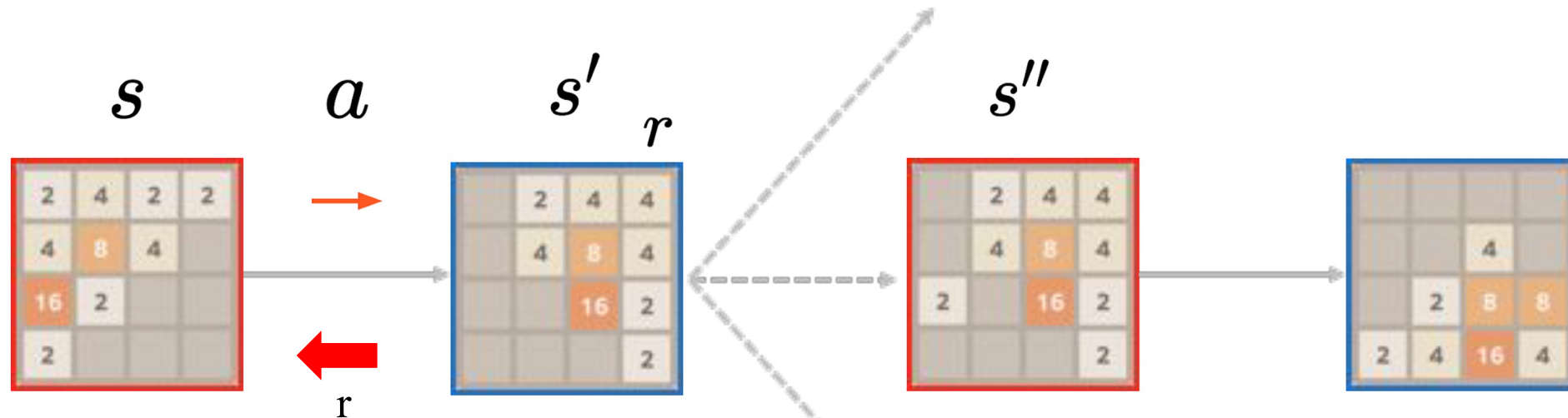
beforestate

afterstate



TD Backup: State

 beforestate
 afterstate



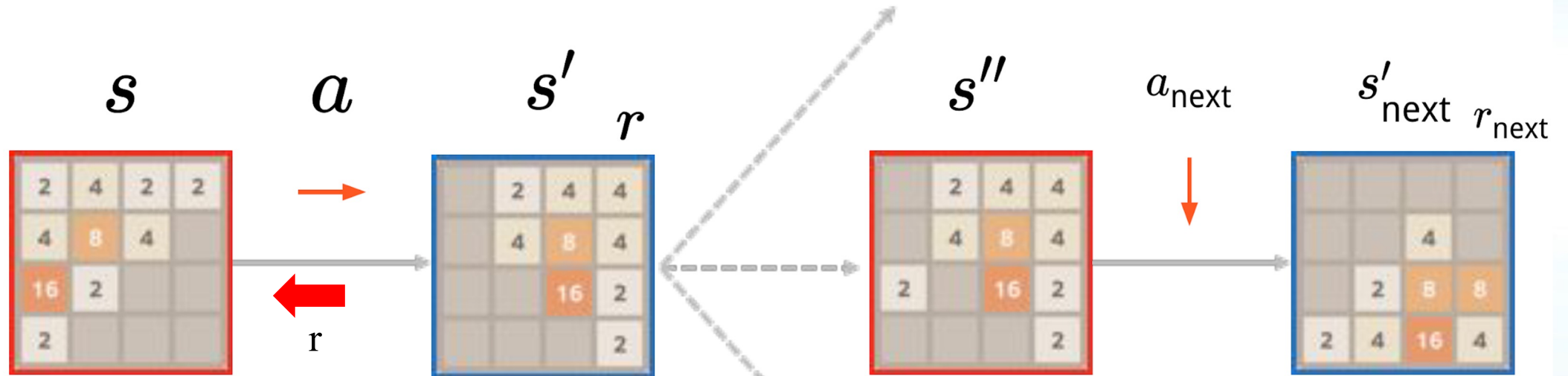
state:



$$V(s) \leftarrow V(s) + \alpha(r + V(s'') - V(s))$$

TD Backup: After-State

beforestate
 afterstate



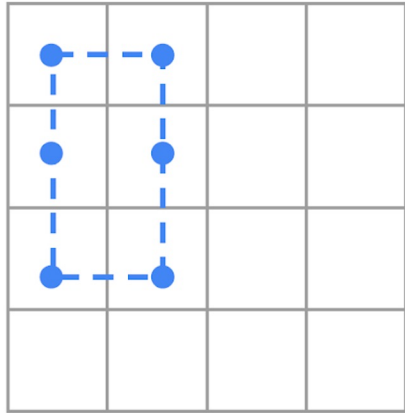
after-state:

$$\begin{array}{c}
 V(s') \qquad \qquad \qquad V(s'_{next}) \\
 \bigcirc \text{-----} \bullet \text{-----} \bigcirc \\
 V(s') \leftarrow V(s') + \alpha(r_{next} + V(s'_{next}) - V(s'))
 \end{array}$$

Why use n-tuple network?

- The expected score/return G_t from a board S
- But, #states is huge
 - About 17^{16} ($=10^{20}$).
 - Empty ($\rightarrow 0$), 2 ($=2^1 \rightarrow 1$), 4 ($=2^2 \rightarrow 2$), 8 ($=2^3 \rightarrow 3$), ..., 65536 ($=2^{16} \rightarrow 16$).
- Need to use value function approximator.

Example: 2048 with n-tuple network

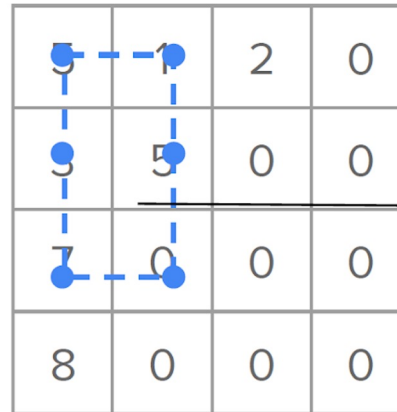


a 6-tuple pattern f_1

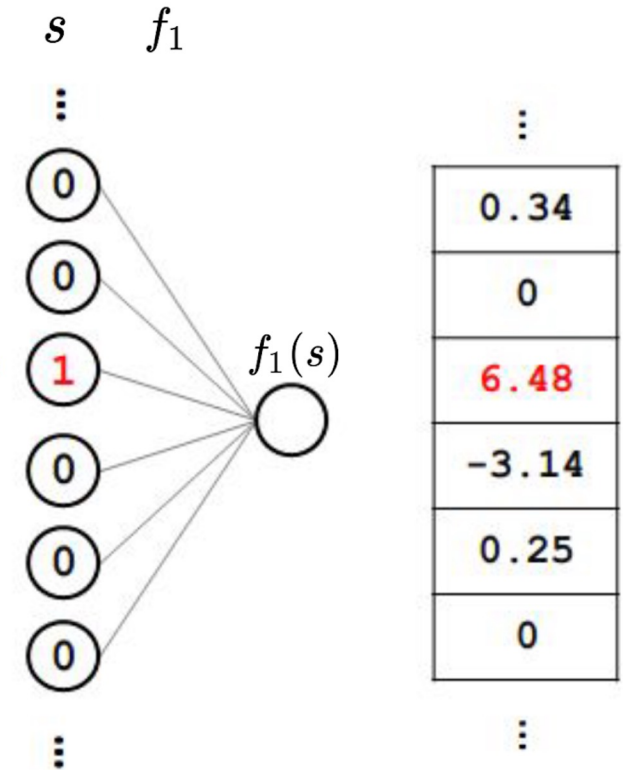


5	1	2	0
3	5	0	0
7	0	0	0
8	0	0	0

board s

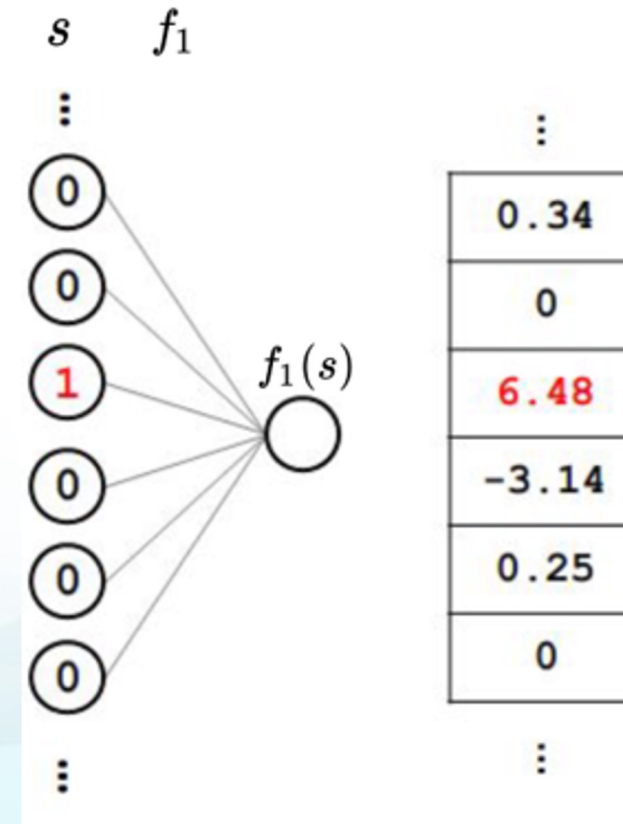
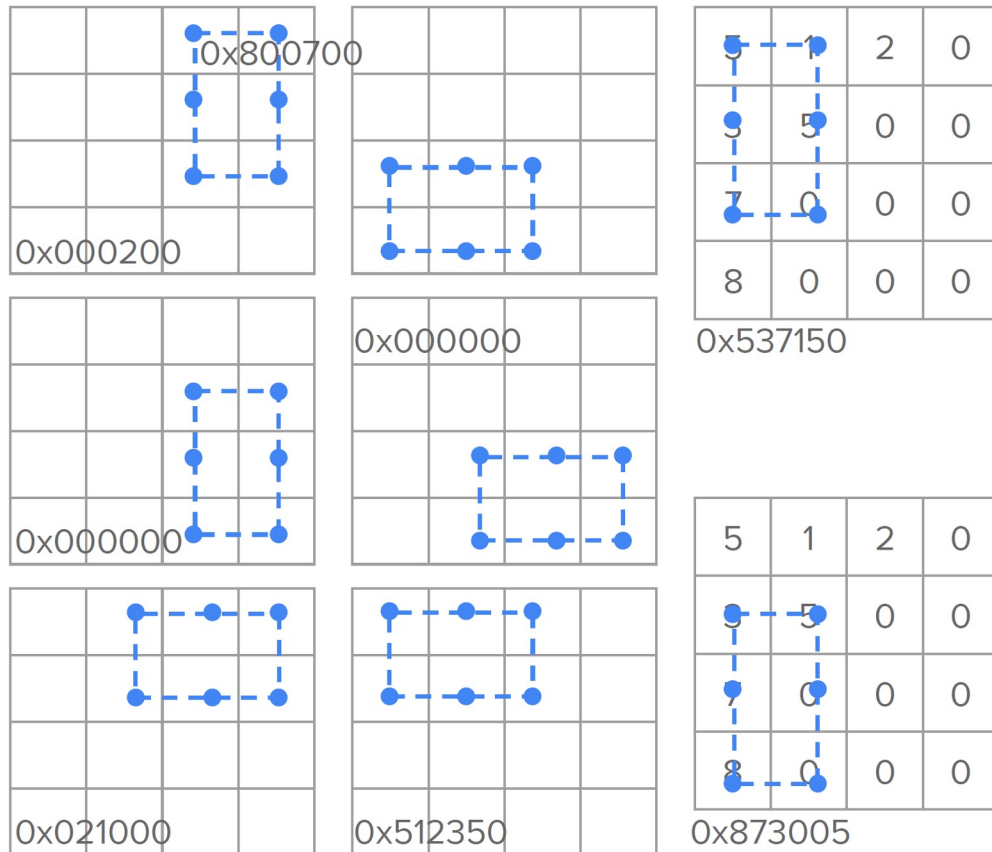


0x537150



All Isomorphism

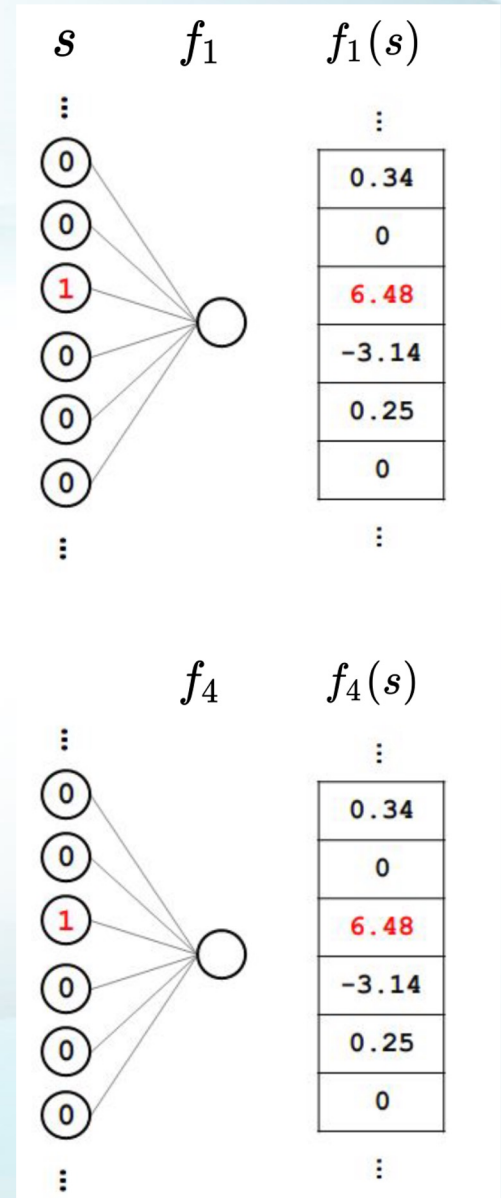
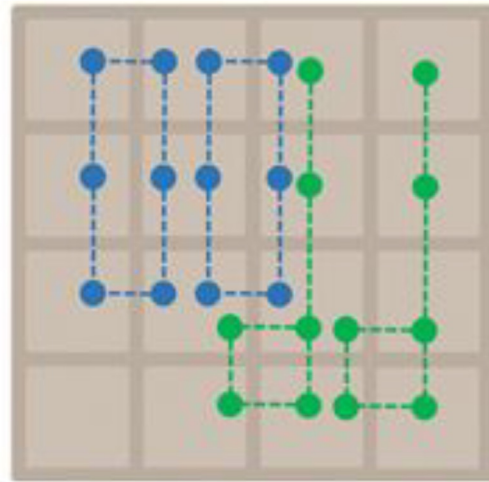
- Rotations and Reflections
- The sum of the eight values can represent the board.



Multiple n-tuple

- Example: 4 kinds of 6-tuple.

$$V(s) = f_1(s) + f_2(s) + f_3(s) + f_4(s)$$



Sample Code

- Implement $V(\text{state})$ and $V(\text{after-state})$ in two files: `td_state.cpp` and `td_afterstate.cpp`
 - Compile with C++11 support
 - `g++ -std=c++11 -O3 -o td_state td_state.cpp`
 - `g++ -std=c++11 -O3 -o td_afterstate td_afterstate.cpp`

Training

:

Save your
model weight

```
// restore the model from file
tdl.load("");

// train the model
std::vector<state> path;
path.reserve(20000);
for (size_t n = 1; n <= total; n++) {
    board b;
    int score = 0;

    // play an episode
    debug << "begin episode" << std::endl;
    b.init();
    while (true) {
        debug << "state" << std::endl << b;
        state best = tdl.select_best_move(b);
        path.push_back(best);

        if (best.is_valid()) {
            debug << "best " << best;
            score += best.reward();
            b = best.after_state();
            b.popup();
        } else {
            break;
        }
    }
    debug << "end episode" << std::endl;

    // update by TD(0)
    tdl.update_episode(path, alpha);
    tdl.make_statistic(n, b, score);
    path.clear();
}

// store the model into file
tdl.save("weights.bin");

return 0;
```

Evaluating (demo):

Set total count
to 1000 games

Load your
model weight

```
int main(int argc, const char* argv[]) {
    info << "TDL2048-Demo" << std::endl;
    learning tdl;

    // set the learning parameters
    float alpha = 0.1;
    size_t total = 1000;
    unsigned seed;
    __asm__ __volatile__ ("rdtsc" : "=a" (seed));
    info << "alpha = " << alpha << std::endl;
    info << "total = " << total << std::endl;
    info << "seed = " << seed << std::endl;
    std::srand(seed);

    // initialize the features
    tdl.add_feature(new pattern({ 0, 1, 2, 3, 4, 5 }));
    tdl.add_feature(new pattern({ 4, 5, 6, 7, 8, 9 }));
    tdl.add_feature(new pattern({ 0, 1, 2, 4, 5, 6 }));
    tdl.add_feature(new pattern({ 4, 5, 6, 8, 9, 10 }));

    // restore the model from file
    tdl.load("weights.bin");

    // train the model
    std::vector<state> path;
    path.reserve(20000);
    for (size_t n = 1; n <= total; n++) {
        board b;
        int score = 0;
```

Scoring Criteria

- Report (30%)
- Demo (70%)
 - The 2048-tile win rate in 1000 games, [winrate₂₀₄₈]. (50%)
 - Questions. (20%)

```
1000    mean = 21355.2    max = 64492
      128      100%      (0.1%)
      256      99.9%      (1.4%)
      512      98.5%      (11.6%)
     1024      86.9%      (51.2%)
     2048      35.7%      (34.6%)
     4096       1.1%      (1.1%)
```

Reminders

- You **can** design your n-tuple.
- You should avoid using CNN in this lab.
- You have to **load your weight** while demo.

References

1. Szubert, Marcin, and Wojciech Jaśkowski. "Temporal difference learning of N-tuple networks for the game 2048." 2014 IEEE Conference on Computational Intelligence and Games. IEEE, 2014.
2. Kun-Hao Yeh, I-Chen Wu, Chu-Hsuan Hsueh, Chia-Chuan Chang, Chao-Chin Liang, and Han Chiang, Multi-Stage Temporal Difference Learning for 2048-like Games, accepted by IEEE Transactions on Computational Intelligence and AI in Games (SCI), doi: 10.1109/TCIAIG.2016.2593710, 2016.
3. Oka, Kazuto, and Kiminori Matsuzaki. "Systematic selection of n-tuple networks for 2048." International Conference on Computers and Games. Springer International Publishing, 2016.
4. moporgic. "Basic implementation of 2048 in Python." Retrieved from Github: <https://github.com/moporgic/2048-Demo-Python> .
5. moporgic. "Temporal Difference Learning for Game 2048 (Demo)." Retrieved from Github: <https://github.com/moporgic/TDL2048-Demo> .
6. lukewayne123. "2048-Framework" Retrieved from Github: <https://github.com/lukewayne123/2048-Framework>